

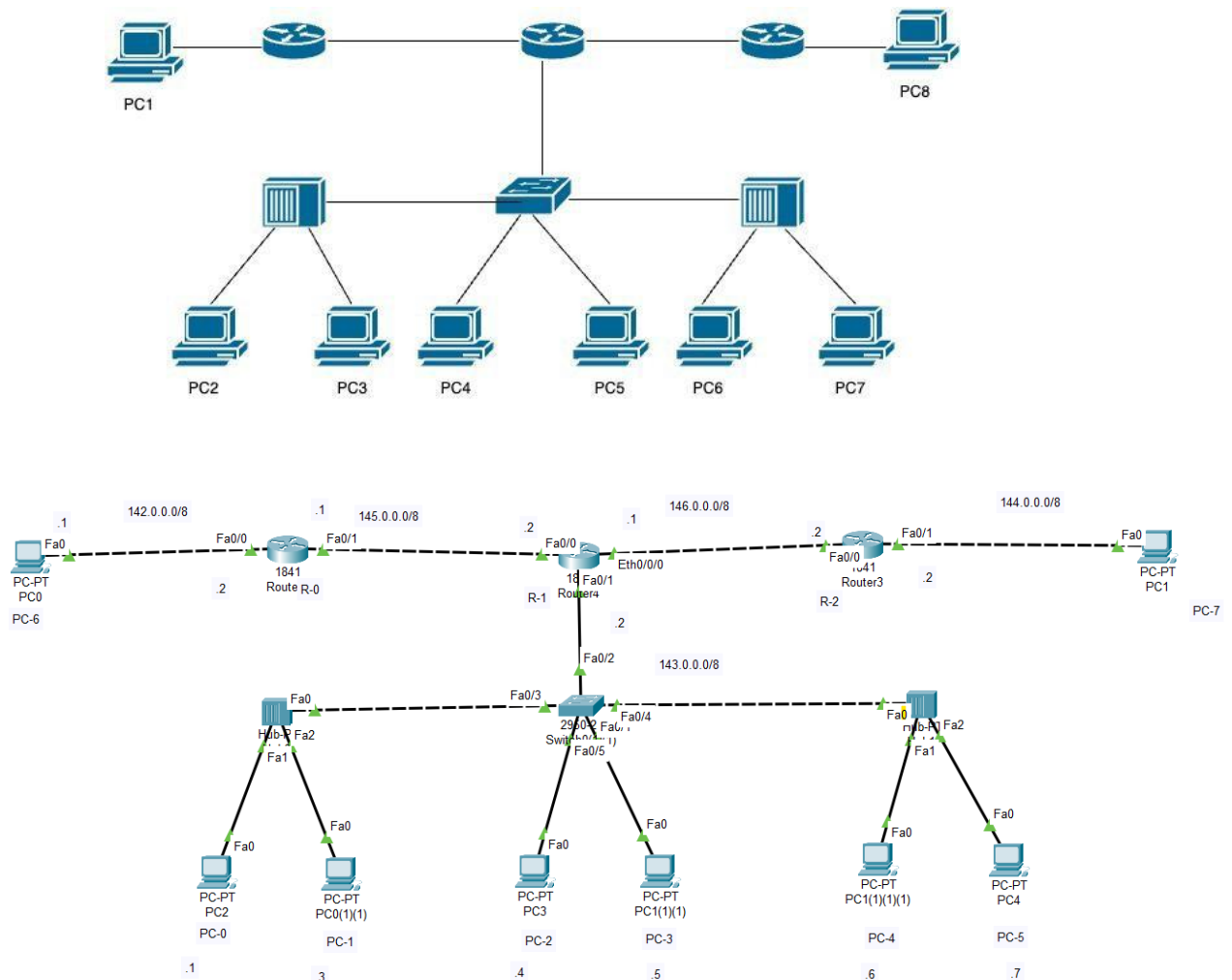
# SET-7

**Note:** While applying IP address, student need to allocate IP address as per his/her student ID. For Example, if student ID is 20ce005 then IP address allocation for first network should start with 5.0.0.0. For subsequent network, it should start with ID+1 i.e. 6.0.0.0, 7.0.0.0. and so on.

**Submission:** After writing answer into this word document, Student need to change name to his ID followed by Set number. Ex 20ce005\_Set1.docx. Upload on assignment segment assigned on teams.

Create this topology in the Cisco Packet Tracer and fill in the tables mentioned below. (10 marks)

**Note:-** Create routing table using static routing protocol.



1. Fill the ARP table entry of all PCs:

PC0

| IP address | MAC Address    |
|------------|----------------|
| 143.1.0.2  | 0000.0ca5.1201 |

PC1

| IP address | MAC Address    |
|------------|----------------|
| 143.4.0.1  | 00e0.a3b4.0e02 |

PC2

| IP address | MAC Address    |
|------------|----------------|
| 143.5.0.1  | 0001.6459.b916 |

PC3

| IP address | MAC Address    |
|------------|----------------|
| 143.5.0.1  | 0001.6459.b916 |

PC4

| IP address | MAC Address    |
|------------|----------------|
| 143.5.0.1  | 0001.6459.b916 |

PC5

| IP address | MAC Address    |
|------------|----------------|
| 142.5.0.1  | 0001.6459.b916 |

PC6

| IP address | MAC Address    |
|------------|----------------|
| 146.5.0.1  | 0001.6459.b916 |

PC7

| IP address | MAC Address    |
|------------|----------------|
| 144.5.0.1  | 0001.6459.b916 |

2. Fill Switch table of all Switches:

| MAC Address     | Ethernet port no |
|-----------------|------------------|
| 0001.6459.B91A6 | Fastethernet 0/1 |
| 0001.C7E0.5A8A  | Fastethernet 0/5 |
| 0060.47C3.A9C7  | Fastethernet 0/3 |
| 00D0.BA2B.9EC6  | Fastethernet 0/3 |

3. Fill Switch table of all hubs:

#### HUB 0

| LINK | Ethernet port no |
|------|------------------|
| UP   | Fastethernet 0   |
| UP   | Fastethernet 1   |
| UP   | Fastethernet 2   |
| DOWN | Fastethernet 3   |
| DOWN | Fastethernet 4   |
| DOWN | Fastethernet 5   |

#### HUB 1

| LINK | Ethernet port no |
|------|------------------|
| UP   | Fastethernet 0   |
| UP   | Fastethernet 1   |
| UP   | Fastethernet 2   |
| DOWN | Fastethernet 3   |
| DOWN | Fastethernet 4   |
| DOWN | Fastethernet 5   |

#### 4. Routing table of all the Routers: (By doing Static entries in routers)

##### Router 0

| Type | Destination Network ID | Next Hop IP address | Hop Count |
|------|------------------------|---------------------|-----------|
| C    | 14.1.0.0/16            | --                  | 0/0       |
| C    | 14.2.0.0/16            | --                  | 0/0       |
| S    | 144.3.0.0/16           | 143.2.0.2           | 1/0       |
| S    | 142.4.0.0/16           | 143.3.0.2           | 1/0       |
| S    | 145.5.0.0/16           | 143.2.0.2           | 1/0       |

Router 1

| Type | Destination Network ID | Next Hop IP address | Hop Count |
|------|------------------------|---------------------|-----------|
| S    | 142.1.0.0/16           | 143.2.0.1           | 1/0       |
| C    | 142.2.0.0/16           | --                  | 0/0       |
| C    | 142.3.0.0/16           | --                  | 0/0       |
| S    | 144.4.0.0/16           | 143.3.0.2           | 1/0       |
| C    | 142.5.0.0/16           | --                  | 0/0       |

Router 2

| Type | Destination Network ID | Next Hop IP address | Hop Count |
|------|------------------------|---------------------|-----------|
| S    | 142.1.0.0/16           | 143.2.0.1           | 1/0       |
| S    | 142.2.0.0/16           | 143.3.0.1           | 1/0       |
| C    | 142.3.0.0/16           | --                  | 0/0       |
| C    | 144.4.0.0/16           | --                  | 0/0       |
| S    | 144.5.0.0/16           | 143.3.0.1           | 1/0       |

**Answer the following questions: (5 marks)**

1. What is the purpose of a default gateway in a network?
2. What is functionality of Hub?
3. What is the difference between static and dynamic routing?
4. Why is the ARP (Address Resolution Protocol) important in networking?
5. What is the next hop IP address?